LECTURE16

28 October 2023 09:37

$\{ x \in \mathbb{Q} : x^2 - 6 \le 0 \}$ $\quad [-\sqrt{6}, \sqrt{6}]$

$(-\sqrt{6}, \sqrt{6})$

Qui $\lim_{n\to\infty} \frac{n^2}{2^n} = 0$ Ratio test

Root test.

concept of order

$$\frac{n+1}{4n^2} < \text{Ques} < \frac{n+1}{n^2}$$

$$\downarrow \qquad \downarrow \qquad \downarrow$$

$$0 \qquad 0 \qquad 0$$

$$\frac{1}{n^2} + \frac{1}{(n+1)^2} + \dots + \frac{1}{(n+m)^2}$$

choose $a_r = \frac{n}{(n+r)^2}$ $\qquad \frac{a_1 + a_2 + \dots + a_n}{n} \to \text{Ques} \to l$

$a_n = \frac{n}{(2n)^2} = \frac{1}{4n} \to 0$

$\sum \frac{n+1}{2n^2 + 7n + 4}$ $\qquad \sum b_n = \sum \frac{1}{n^2}$ $\qquad$ Qui $\frac{a_n}{b_n} \neq 0$ finite

$\sum \frac{n^n}{n!} x^n = ex < 1$ cgt $\quad x < 1/e$ cgt

$ex > 1$ div $\quad x > 1/e$ div

Conti: $x^2 \cos \frac{1}{x}$ at $x = 0$ & $x = \frac{1}{\pi}$

$\frac{1}{\pi^2} \cos \pi$ = finite $\qquad \cos \frac{1}{x}$ at $\frac{1}{\pi}$

0

$\sum a_n x^n$ about 0

$\sum a_n (x-c)^n$ about 'c'

$\sum a_n x$ about 0

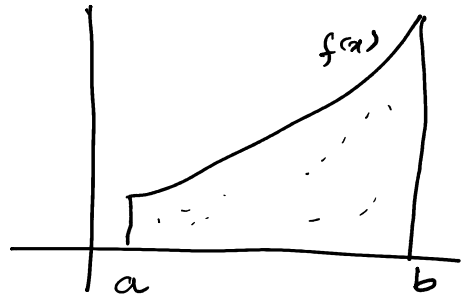
$\sum a_n (x-c)^n$ about 'c'

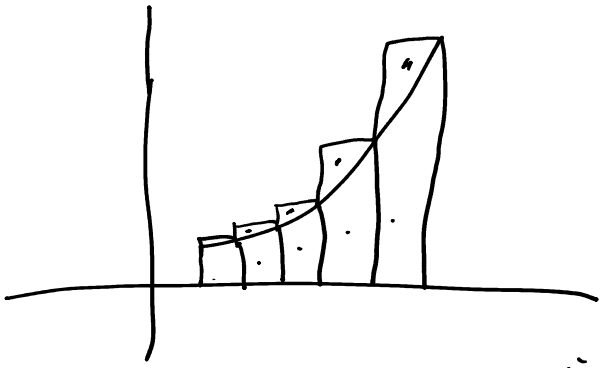
$x \to x_0$

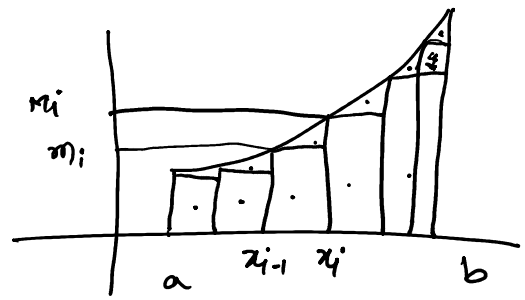
$\sum \underbrace{a_n (x_0 - c)^n}_{\sum b_n}$

Integration $\int_a^b f(x) =$

$\sum_{n=1}^{\infty} f_n = \int_0^1$







Upper Sum $= \sum M_i (x_i - x_{i-1})$

Lower Sum $= \sum_{i=1}^{n} m_i (x_i - x_{i-1})$

Lower Sum = Upper Sum

$L(P, f) = U(P, f)$

$$\int_{\bar{a}}^{b} f(x)\,dx = \int_{a}^{\bar{b}} f(x)\,dx$$